

Research in Information Security

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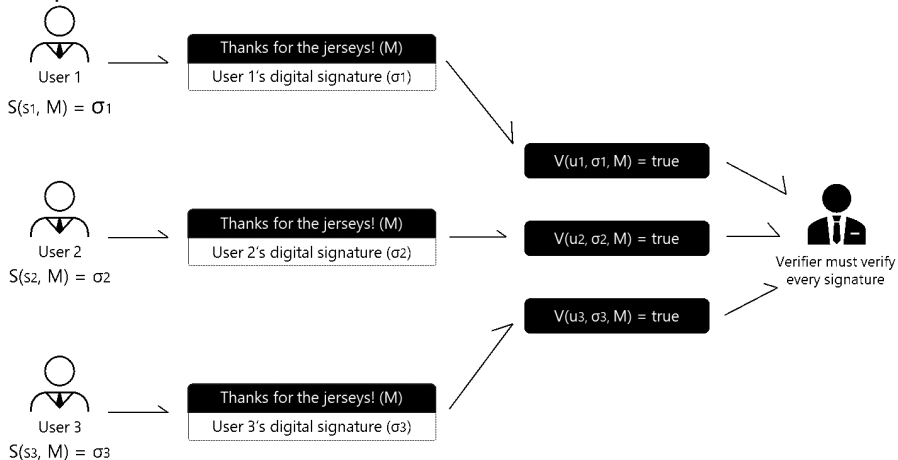
Algorithms and Approaches of Proxy Signatures

Digital Signature

- Digital signature is a cryptographic means through which the authenticity, data integrity and signer's non-repudiation can be verified.
- Typically, digital signature of a document is a piece of information encrypted by the signer's private key.
- Numerous researches have shown significant contributions to this field using various cryptographic primitives.
- Nevertheless, there are many practical environments where digital signatures do not possess specific requirements, and thereby digital signatures appear in several other forms, namely proxy signatures, multi signatures, blind signatures, ring signatures etc.

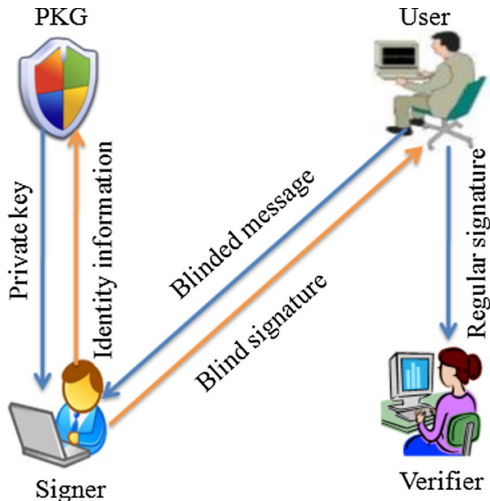
Multi-signature

- A multi-signature scheme enables a group of signers to produce a compact, joint signature on a common document, and has many potential uses.

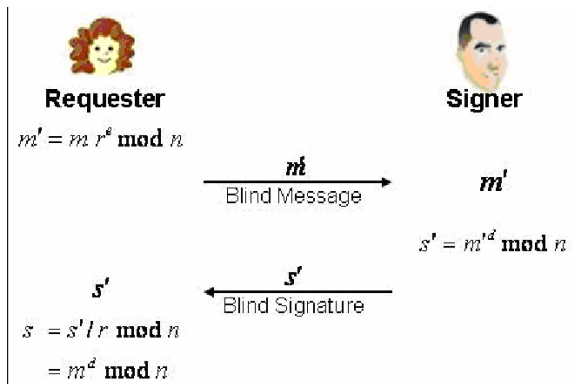


Blind signature

- In many applications involving anonymity, it is desirable to allow a participant to sign a message without knowing what the message is. This is called a blind signature.

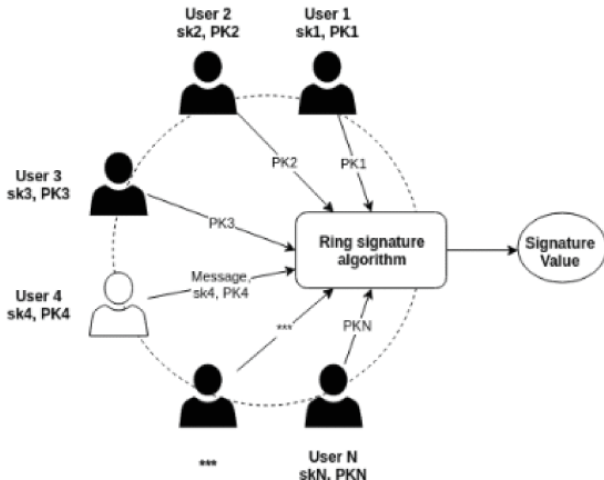


- Bleumer, G. (2011). Chaum Blind Signature Scheme. In: van Tilborg, H.C.A., Jajodia, S. (eds) Encyclopedia of Cryptography and Security. Springer, Boston, MA.



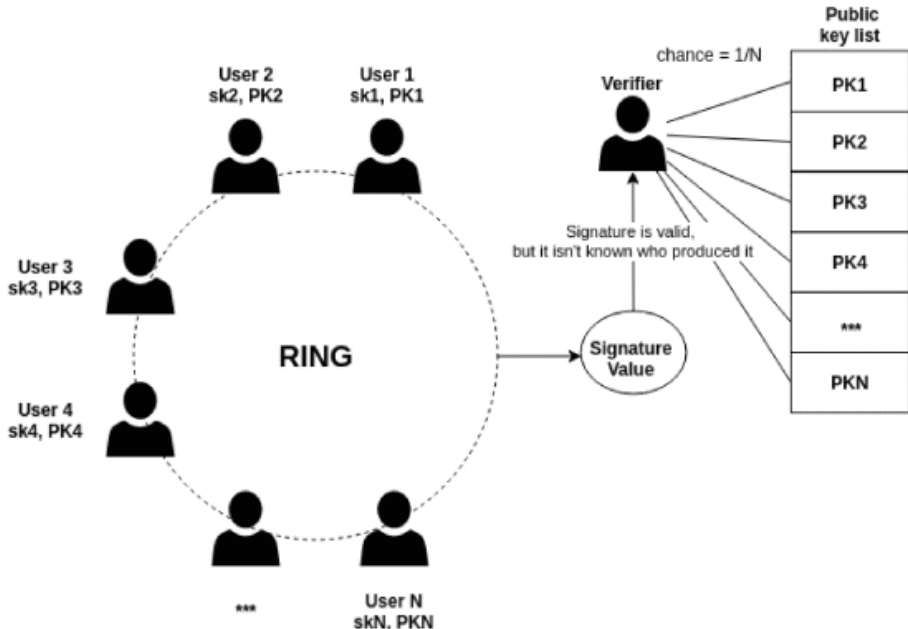
Ring signature

- Ring signatures, first introduced by Rivest, Shamir, and Tauman, enable a user to sign a message so that a ring of possible signers (of which the user is a member) is identified, without revealing exactly which member of that ring actually generated the signature.



Signature calculation process

Ring signature verification process



Aggregate Signature

- In a general aggregate signature scheme, signatures are generated by individual users. They can then be combined into an aggregate signature by some aggregating party.
- An aggregate signature is the same length as an ordinary signature in the underlying scheme.

